DETECTING ADVERSE DRUG REACTIONS IN TEXT

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Abstract

Official recommendations surrounding adverse drug reactions are largely curated from independent reports or electronic health records submitted by drug manufacturers and clinicians. Such reports are time-consuming and tedious to label, making adverse drug reactions an under-reported and inefficient field to study. Hence, this project seeks to build a system using natural language processing techniques to automatically crawl through texts and identify specific adverse drug reactions and drug associations.

Introduction

A recent study quantifying the frequency of adverse drug reactions (ADRs) suggests that as many as one in twenty patients requiring emergency hospitalization in the United States was admitted due to ADRs (Komagamine, 2024). This statistic underscores the prevalence of ADRs in the population and the importance of identifying and treating them. However, the study of ADRs is made difficult with the issue of underreporting and understudied drug associations. Health professionals such as medical doctors or pharmacists typically report ADR occurrences in the field to pharmaceutical companies and manufacturers, which then pass on reports to governmental agencies to collate important information. Although some countries have developed processes that aid in the report of ADRs, like the Yellow Card Scheme in the United Kingdom (Medicines & Healthcare products Regulatory Agency, n.d.), reporting is still not a mandatory process for many countries. Coupled with a lack of standardized documentation for reporting, the largely unregulated process of monitoring and information dissemination in reports makes it difficult to identify specific ADR and drug associations for research. Thus, the development of an automated system to extract ADR and drug pairs in texts will go a long way in aiding medical intervention and treatment development for ADRs.

Literature Review

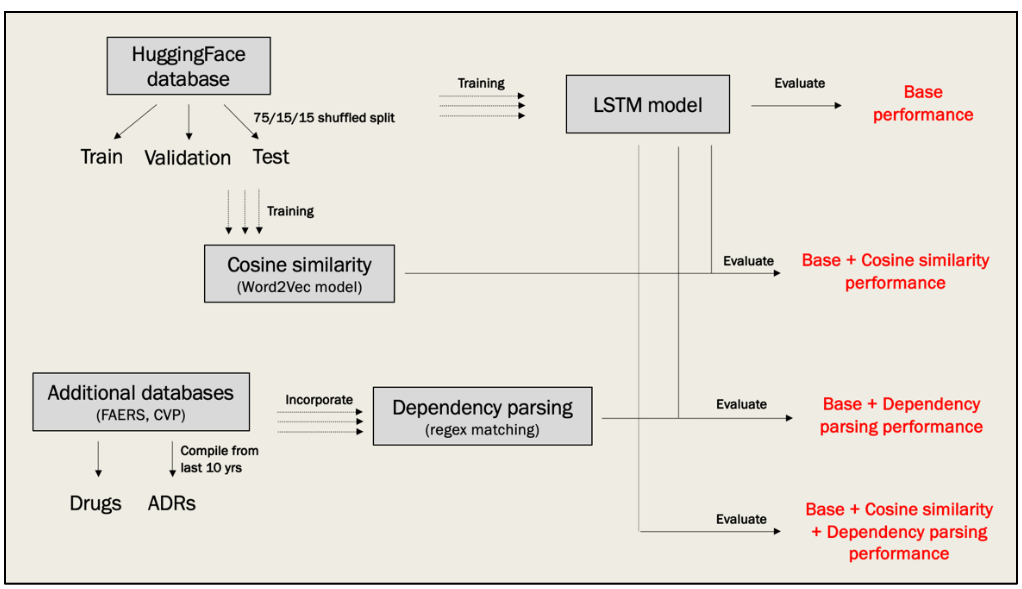
Significant work has been done on detecting ADRs in texts. Burgt et al. developed a text mining algorithm to identify ADRs in Dutch EHRs (Burgt et al., 2024), combining R-scripts and public databases such as Medical Dictionary (MEDRA) for Regulatory Activities and Systematized Nomenclature of Medicine Clinical Terms (NOMED-CT) to determine related terms to find. Their algorithm showed a sensitivity of 73% and a positive predictive value of 70% in identifying ADRs. Other researchers such as Zhang et al. have looked at extracting ADRs on social media in an attempt to boost under-reported ADRs and expand the types of sources we can use to obtain information on them (Zhang et al., 2020). Integrating shallow and deep linguistic features to train models, they found that their novel approach outperformed other state-of-the-art methods with an AUC of 94.44% and 88.97% on two datasets. Some teams like Bollegalla et al. have relied on just lexical patterns alone to parse ADR and drug events, mining social media texts to find a causality-sensitive approach to represent them (Bollegala et al., 2017). By framing drug and ADR events as causality instances instead of co-occurring instances, the team was able to detect ADRs with an accuracy of 74%.

Methodology Overview

This project has developed an automated system combining natural language processing (NLP) techniques to identify a set of ADR and associated drugs in English texts. The final prediction system consists of a bi-directional Long Short-Term Memory (LSTM) model to generate base predictions, then utilizes cosine similarity scores and dependency parsing to improve and make final predictions. The prediction system was designed and evaluated in 4 phases to assess the performance of each NLP piece (Figure 1). In the first phase, a LSTM model was trained and evaluated to assess baseline prediction performance. The second and third phase consists of utilizing either cosine similarity scores or dependency parsing to improve upon LSTM model predictions, then evaluating the performance of the two prediction systems respectively. The last phase combines all three NLP pieces to form and evaluate the performance of the final prediction system.

Figure 1

Methodology overview of final system design and evaluation



Datasets used for training and evaluation

Training, validation and testing datasets used across all 4 evaluation phases were extracted from the ade_corpus_v2 benchmark dataset found on the HuggingFace database. The benchmark dataset provides a dataframe of drug, ADR, text containing the drug and ADR listed, and the start and end character indexes of the drug and ADR pair found in the text. The dataset was shuffled and split into training, validation and testing datasets using a 70%, 15% and 15% proportion. The final datasets contained 4774, 1023 and 1023 rows of data for training, validation and testing respectively.

Additionally, a list of standard drug and ADR terms were extracted from reports publicly available in the FDA Adverse Event Reporting System (FAERs) and Canada Vigilance Program (CVP) databases for usage in Phases 3 and 4. FAERs is a database containing information related to ADR and medication error reports submitted to the U.S. Food and Drug Administration, and contains terms that adhere to the MEDRA terminology. The CVP is a program administered by the government of Canada to collect and report on potential ADRs associated to health products sold in Canada, which includes products such as prescription and non-prescription medications. This project extracted terms from reports submitted in the last 10 years (2015-2025), lowercase normalized the terms and retained only unique, complete drug and ADR pairs. The final list of terms consists of 59,681 drugs and 16,824 ADRs.

Phase 1: Long Short-Term Memory Model Training

A standard bi-directional LSTM sequence tagger model was implemented using the FLAIR module to generate base ADR and drug predictions from input texts. The training and validation datasets extracted from HuggingFace were used to train the model, using a learning rate of 0.1 and batch size of 3 across 10 epochs as training parameters. To preprocess datasets for FLAIR training, texts in the datasets were tokenized, sequence tagged with 5 IOB tags and converted into FLAIR-compatible text files. The 5 IOB tags used are listed alongside their descriptions in Table 1.

Table 1

Table of IOB tags and tag descriptions used to sequence tag HuggingFace datasets

IOB Tag	Description
B-DRUG	Tags "Beginning-Drug" tokens. Word token at beginning of a drug entity span.
I-DRUG	Tags "Intermediate-Drug" tokens. Word token that occurs within a drug entity span.
B-ADR	Tags "Beginning-ADR" tokens. Word token at beginning of an ADR entity span.
I-ADR	Tags "Intermediate-ADR" tokens. Word token that occurs within an ADR entity span.
O	Tags "Others" tokens. Word token that does not belong to a drug or ADR entity span.

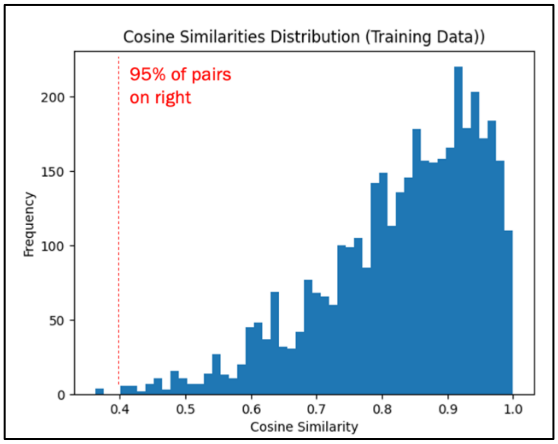
Phase 2: Incorporating and Training Cosine Similarity Score with LSTM System

Cosine similarity score is a similarity measurement of word vectors that represent drug and ADRs. Cosine similarity scores were utilized in phase 2 to improve false positives generated by the LSTM model. The prediction system implemented consists of using the LSTM model trained in Phase 1 to generate drug and ADR predictions, then use a trained word2vec model to generate a cosine similarity score for each pair. Final drug and ADR pairs generated by the prediction system were retained based on a minimum cosine similarity score threshold. A pretrained word2vec "en_core_web_sm" model was implemented using the spaCy module, then further trained using the HuggingFace training dataset. The training dataset was first lowercase normalized, then tokenized to train the word2vec model using a vector size of 100, window of 5, min_count of 1, sg of 1 and negative of 8.

To determine an appropriate cosine similarity score threshold for discarding false positives, cosine similarity scores were generated by the word2vec model for all drug and ADR pairs found in the HuggingFace training dataset. The minimum threshold to detect a true pair is then identified by finding the minimum score for at least 95% of the pairs, where 95% was chosen to avoid overfitting to the training dataset. A minimum threshold of 0.4 cosine similarity score was used in the prediction system, since 95% of all pairs obtained a score above that threshold (Figure 2).

Figure 2

Histogram showing cosine similarity scores of drug and ADR pairs in HuggingFace training dataset



Phase 3: Incorporating and Training Dependency Parsing with LSTM System

Many teams much like Bollegalla et al. have demonstrated the feasibility of identifying drug and ADR terms using lexical patterns. In this prediction system, a simple dependency parsing method of identifying pairs using causal effect terms is implemented. Drug and ADR terms are often expressed in the same sentence alongside words such as "resulting" or "due to". For instance, an example text containing a drug and ADR pair found in the HuggingFace training dataset is as follows "the fatal stroke may have resulted from arterial spasm caused by ergotamine overdosage". Here, the lexical pattern is demonstrated by the drug term "ergotamine" being associated to the ADR term "stroke" by the causal effect term "resulted from". Thus, dependency parsing can be utilized to generate alternative predictions besides those made by the LSTM model, potentially improving detection of true positives.

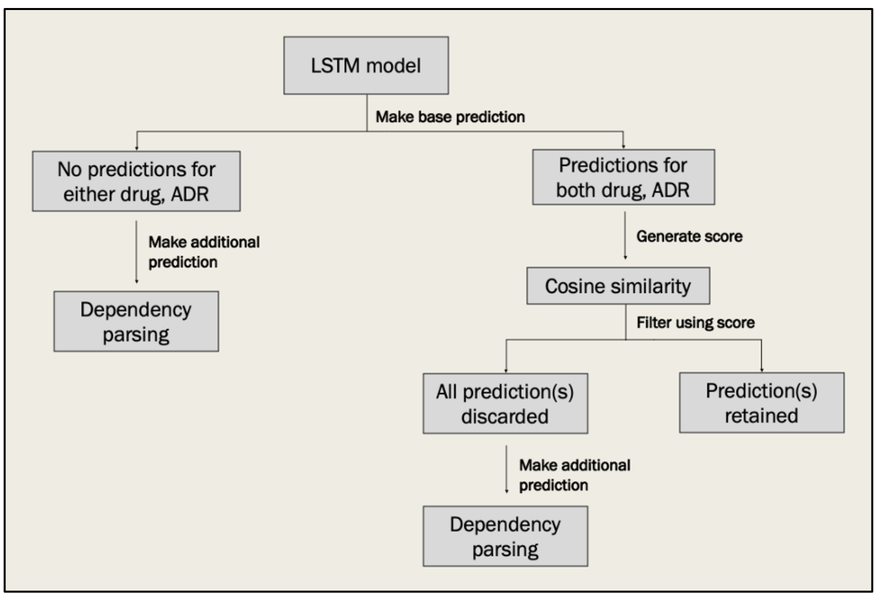
In Phase 3, the prediction system implemented consists of using the LSTM model trained in Phase 1 to generate drug and ADR predictions, then using dependency parsing to generate alternative predictions if the LSTM model did not make one for either drug or ADR. Dependency parsing consists of a 2-step process. First, sentences containing potential drug and ADR pairs are identified using a simple regex pattern of *"* Noun Phrase * causal term * Noun Phase *"*. A list of 29 causal terms, including variations of terms such as "result", "results", "resulting" are used for regex pattern matching. Since drugs and ADRs manifest as noun phrases in a sentence, but not all noun phrases are necessarily drugs or ADRs, the second step compares all extracted noun phrases with the list of FAERs and CVP drug and ADR terms to identify potential drug and ADR pairs. The final predictions generated by the prediction system thus consists of LSTM model base predictions and alternative dependency parsing predictions.

Phase 4: Final Prediction System Training

The final prediction system was implemented in phase 4, which consists of using the LSTM model trained in Phase 1 to generate drug and ADR predictions, then incorporating cosine similarity and dependency parsing to improve upon base predictions. Cosine similarity was used to discard false positives in the same manner as in Phase 2 and utilized when the LSTM model made a complete drug and ADR pair prediction. If the LSTM model fails to predict a complete pair, or if all pair predictions were discarded as a result of low cosine similarity scores, dependency parsing was used to generate alternative drug or ADR predictions in the same manner as in Phase 3. The steps used by the final prediction system to generate drug and ADR pairs from texts are shown in Figure 3.

Figure 3

Steps carried out by final prediction system to predict drug and ADR pairs in text



Evaluation and Results

The systems were evaluated in each of the 4 phases for prediction performance using standard precision, recall and F1 score metrics. All phases used the HuggingFace testing dataset for evaluation. The precision metric measures the fraction of relevant instances among retrieved instances, of which a higher value indicates a higher number of true drug and ADR pairs being predicted than false ones. The recall metric measures the fraction of relevant retrieved instances over all relevant instances, of which a higher value indicates a higher proportion of possible true drug and ADR pairs being predicted. The F1 score metric combines the precision and recall metric and weights them equally, and can be interpreted as a measurement of the overall predictive performance. The evaluation results obtained in each of the 4 phases for drug, ADR, and overall detection are listed in Tables 2, 3, 4 and 5.

Table 2

Evaluation results obtained for drug, ADR and overall detection in Phase 1

Metric	Precision	Recall	F1
Drug	0.714	0.864	0.782
ADR	0.654	0.788	0.715
Overall	0.691	0.834	0.756

Table 3

Evaluation results obtained for drug, ADR and overall detection in Phase 2

Metric	Precision	Recall	F1
Drug	0.718	0.581	0.642
ADR	0.654	0.535	0.589
Overall	0.688	0.560	0.617

Table 4

Evaluation results obtained for drug, ADR and overall detection in Phase 3

Metric	Precision	Recall	F1
Drug	0.713	0.868	0.783
ADR	0.597	0.810	0.684
Overall	0.664	0.841	0.742

Table 5

Evaluation results obtained for drug, ADR and overall detection in Phase 4

Metric	Precision	Recall	F1
Drug	0.713	0.868	0.783
ADR	0.597	0.801	0.684
Overall	0.664	0.841	0.742

The baseline performance of the prediction system in Phase 1 was reasonably high given the parameters set for LSTM model training. The LSTM model obtained lower scores for ADR detection compared to drug detection, suggesting that the system struggled more with ADR than drug detection in texts. A lower overall precision score compared to recall also indicates a poorer performance in keeping false drug and ADR predictions low as opposed to detecting true pairs.

With the incorporation of cosine similarity scores in phase 2, a higher precision was obtained for drug detection but the same precision was obtained for ADR detection. This indicates more drugs were correctly detected when using cosine similarity scores, without compromising on ADR detection performance. However, a steep drop in recall was recorded across the categories in this phase, indicating a lower proportion of true positives were detected. This is expected, since base predictions were discarded without replacement with alternative ones. A lower recall score might also suggest that the minimum score threshold is set too high, resulting in some true positive cases being incorrectly discarded.

Using dependency parsing to generate alternative predictions to supplement LSTM model predictions proved useful in phase 3, obtaining higher precision and recall for majority of the detection categories. The prediction system only saw a marginal drop in recall for ADR detection, indicating that dependency parsing made more false ADR predictions compared to the base prediction system.

Lastly, incorporating both cosine similarity and dependency parsing to supplement LSTM model predictions in Phase 4 resulted in similar prediction performance as the baseline performance. The final prediction system only showed a slight drop in recall for ADR detection when compared to the base system. Upon deeper investigation, it was found that the final prediction system generated many true drug and ADR pairs that were not labelled in the HuggingFace testing dataset. For instance, the following text "interferon alpha induced polymyositis and cardiomyopathy is diagnosed" appeared in the testing dataset with "interferon alpha" labelled as the drug and "cardiomyopathy" labelled as the ADR. However, the final prediction system predicted the correct drug and ADR labels, alongside "polymyositis" as a second ADR term. Due to multiple instances of incomplete drug and ADR pairs in the testing dataset, many true pairs predicted by the final prediction system were misclassified as false positives instead. The removal of true positives by cosine similarity scores, alongside the generation of more true and false positives by dependency parsing, likely explains the lack of performance gain in the final prediction system compared to the base system.

Conclusion

This project has explored and evaluated some of the NLP methods that can be used for ADR and drug detection in texts. With limitations in hardware, there are additional improvements the project can explore in the future to build a better ADR and drug detection system. One possible improvement is to utilize drug and ADR pairs extracted from the FAERs and CVP databases to obtain abstracts from scientific literature using the PubMED API. Parsed abstracts, drug and ADR terms can then be used as additional training data to supplement the HuggingFace datasets for LSTM model and word2vec model training in Phases 1, 2 and 4. Another possible improvement is expanding the list of drug and ADR terms extracted from FAERs and CVP by querying additional public databases, potentially adding more alternative terms including acronyms into the list. With a bigger reference list, more drug and ADR pairs can be correctly identified in Phases 3 and 4. Lastly, it is also worthwhile to explore increasing the number of epochs or batch size and decreasing the learning rate when training the LSTM model in Phase 1. Tweaking the training parameters in this manner might allow the LSTM model to learn more useful patterns across more training iterations for drug and ADR detection.

Project Code

Datasets and code used in the project is publicly available on GitHub via https://github.com/Amandahsr/AdverseDrugReactionInText.git.

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